

II-Probabilistic Approaches Independence and Graphical Models

Foundations of Neuro-Symbolic AI

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Outline

- ▶ Independence of variables
- ▶ Conditional independence
- ▶ Decompositions given independence
- ▶ Bayesian networks
- ▶ Markov networks

How to beat the curse of dimensions?

Probability distribution of factored systems with states $\times_{k \in [d]} [m_k]$ has

$$\left(\prod_{k \in [d]} m_k \right) - 1$$

degrees of freedom (coordinates to specify and store).

Mitigation: [Tensor network decompositions](#)

Idea

Exploit the "dependence" structure of variables to find decompositions of probability distributions.

Example: At the dentist

Consider the joint distribution of the variables X_0 (Toothache), X_1 (Cavity), X_2 (Catch), with a variable X_3 , denoting whether there is a Cloud outside the dentists lab.

- ▶ We now have an order-4 probability tensor $\mathbb{P}[X_0, X_1, X_2, X_3]$
- ▶ Intuitively, knowing Cloud should not affect the probability of having a cavity, so why shall we care?

To be more precise let us consider the distribution:

$$\mathbb{P}[X_0, X_1, X_2, X_3] = \begin{pmatrix} 0.288 & 0.004 \\ 0.032 & 0.006 \end{pmatrix} \begin{pmatrix} 0.077 & 0.036 \\ 0.008 & 0.054 \end{pmatrix} [X_0, X_1, X_2, X_3]$$

Such redundancies are captured by the concept of independence.

Formal definition of Independencies

Definition (Independence)

Given a joint distribution $\mathbb{P}[X_{\mathcal{V}}]$ and two disjoint subsets $A, B \subset \mathcal{V}$, we say that the variables X_A are independent from the variables X_B if the marginal distributions $\mathbb{P}[X_{A \cup B}]$, $\mathbb{P}[X_A]$ and $\mathbb{P}[X_B]$ satisfy for any index tuples x_A, x_B

$$\mathbb{P}[X_A = x_A, X_B = x_B] = \mathbb{P}[X_A = x_A] \cdot \mathbb{P}[X_B = x_B] .$$

We denote in this case $(X_A \perp X_B)$

Interpretation: Knowing the value of variables B (and conditioning on them) does not change the distribution of variables A .

Contraction criterion for independence

Theorem

If and only if the X_A are independent from the variables X_B with respect to a distribution $\mathbb{P}[X_V]$ then

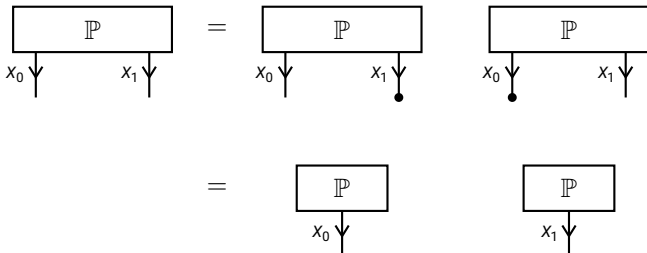
$$\langle \mathbb{P}[X_{[d]}] \rangle_{[X_A, X_B]} = \langle \mathbb{P}[X_{[d]}] \rangle_{[X_A]} \otimes \langle \mathbb{P}[X_{[d]}] \rangle_{[X_B]} .$$

In a tensor network diagram this is the decomposition:

The diagram illustrates the decomposition of a joint distribution into a tensor product of marginal distributions. On the left, a box labeled $\langle \mathbb{P}[X_{[d]}] \rangle_{[X_A, X_B]}$ has four vertical arrows pointing downwards. The first two arrows are grouped under the label X_A and the next two under X_B . This is followed by an equals sign. On the right, there are two separate boxes. The first box is labeled $\langle \mathbb{P}[X_{[d]}] \rangle_{[X_A]}$ and has two vertical arrows pointing downwards, labeled X_A . The second box is labeled $\langle \mathbb{P}[X_{[d]}] \rangle_{[X_B]}$ and has two vertical arrows pointing downwards, labeled X_B .

Storage of distributions by marginal distributions

Independence allows the decomposition into tensor products



Exponential to linear storage demand

Instead of storing $m_0 \cdot m_1$ coordinates, we can store $\mathbb{P}$ in the case of independency with $m_0 + m_1$ demand.

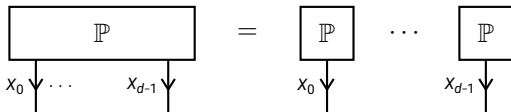
Decomposition of pairwise independent variables

Imagine now that we have a distribution $\mathbb{P} [X_{[d]}]$, such that for any $k, \tilde{k} \in [d]$

$$(X_k \perp X_{\tilde{k}})$$

We then have the decomposition:

$$\mathbb{P} [X_{[d]}] = \bigotimes_{k \in [d]} \mathbb{P} [X_k]$$



Exponential to linear storage demand

Instead of storing $\prod_{k \in [d]} m_k$ coordinates, we can store $\mathbb{P}$ using $\sum_{k \in [d]} m_k$ coordinates of the marginal distributions.

Outline

- ▶ Independence of variables
- ▶ **Conditional independence**
- ▶ Decompositions given independence
- ▶ Bayesian networks
- ▶ Markov networks

Assumption of independence

Independence is a very restrictive assumption:

- ▶ We need dependence of effects on causes for machine learning to work.
- ▶ Independence covers only the most extreme form of tensor network decompositions.

Example: In the dentist example (without the **Cloud** variable), no variables are independent from each other.

Are there more common properties corresponding to more generic tensor decompositions?

Formal definition of Conditional Independencies

Definition (Conditional Independence)

Given a joint distribution of variables X_A , X_B and X_C , we say that X_A and X_B are independent given X_C if for any index tuple (x_A, x_B, x_C)

$$\mathbb{P}[X_A = x_A, X_B = x_B | X_C = x_C] = \mathbb{P}[X_A = x_A | X_C = x_C] \cdot \mathbb{P}[X_B = x_B | X_C = x_C],$$

We denote this situation by

$$(X_A \perp X_B) \mid X_C.$$

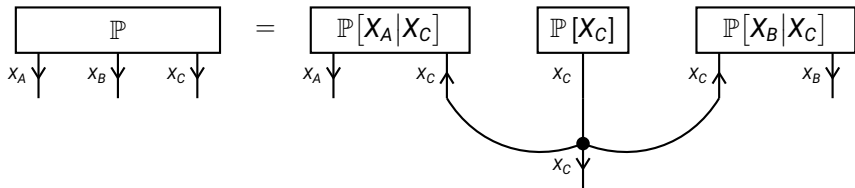
Interpretation: Conditional independence means that X_A and X_B are independent with respect to any slice $\mathbb{P}[X_A, X_B | X_C = x_C]$.

Conditional independence as a contraction equation

We have $(X_A \perp X_B) \mid X_C$ if and only if

$$\mathbb{P}[X_A, X_B \mid X_C] = \langle \mathbb{P}[X_A \mid X_C], \mathbb{P}[X_B \mid X_C] \rangle_{[X_A, X_B, X_C]} .$$

We depict conditional independence by tensor network decompositions:



Example: Being at a dentist

Task: Let us investigate, whether **Toothache** (X_0) and **Catch** (X_2) are independent given **Cavity** (X_1), with respect to the distribution:

$$\mathbb{P}[X_0, X_1, X_2] = \begin{pmatrix} 0.576 & 0.008 \\ 0.064 & 0.012 \end{pmatrix} \begin{pmatrix} 0.144 & 0.072 \\ 0.016 & 0.108 \end{pmatrix} [X_0, X_1, X_2]$$

For $X_1 = 0$:

For $X_1 = 1$:

Example: Being at a dentist

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For $X_1 = 0$:

$$\mathbb{P}[X_0, X_2 | X_1 = 0] = \begin{pmatrix} 0.72 & 0.08 \\ 0.18 & 0.02 \end{pmatrix} [X_0, X_2]$$

For $X_1 = 1$:

Example: Being at a dentist

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For $X_1 = 0$: **Independent** ✓

$$\mathbb{P}[X_0, X_2 | X_1 = 0] = \begin{pmatrix} 0.72 & 0.08 \\ 0.18 & 0.02 \end{pmatrix} [X_0, X_2] = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix} [X_0] \otimes \begin{pmatrix} 0.9 \\ 0.1 \end{pmatrix} [X_2]$$

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For $X_1 = 1$: **Independent** ✓

$$\mathbb{P}[X_0, X_2 | X_1 = 1] = \begin{pmatrix} 0.04 & 0.36 \\ 0.06 & 0.54 \end{pmatrix} [X_0, X_2]$$

Example: Being at a dentist

Task: Let us investigate, whether **Toothache** (X_0) and **Catch** (X_2) are independent given **Cavity** (X_1), with respect to the distribution:

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For $X_1 = 1$: **Independent** ✓

$$\mathbb{P}[X_0, X_2 | X_1 = 1] = \begin{pmatrix} 0.04 & 0.36 \\ 0.06 & 0.54 \end{pmatrix} [X_0, X_2] = \begin{pmatrix} 0.4 \\ 0.6 \end{pmatrix} [X_0] \otimes \begin{pmatrix} 0.1 \\ 0.9 \end{pmatrix} [X_2]$$

Example: Being at a dentist

Task: Let us investigate, whether **Toothache** (X_0) and **Catch** (X_2) are independent given **Cavity** (X_1), with respect to the distribution:

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Conclusion: The conditional independence holds, that is $(X_0 \perp X_2) | X_1$.

Outline

- ▶ Independence of variables
- ▶ Conditional independence
- ▶ **Decompositions given independence**
- ▶ Bayesian networks
- ▶ Markov networks

Chain Rule: Decomposing Probabilities

Theorem (Chain Rule)

For any joint probability distribution $\mathbb{P}[X_{[d]}]$ we have

$$\mathbb{P}[X_{[d]}] = \langle \{\mathbb{P}[X_k | X_{[k]}] : k \in [d]\} \rangle_{[X_{[d]}]} .$$

Proof sketch: We apply the Bayes theorem to get

$$\mathbb{P}[X_{[d]}] = \langle \mathbb{P}[X_0], \mathbb{P}[X_1, \dots, X_{d-1} | X_0] \rangle_{[X_{[d]}]} .$$

and further apply the Bayes theorem iteratively on the conditioned distribution.

Problem: The tensors $\mathbb{P}[X_k | X_{[k]}]$ are of order $d + 1$!

Can we smallen the order based on independence assumptions?

Simplification of conditional distributions

Theorem

When $(X_A \perp X_B) \mid X_C$ then

$$\mathbb{P}[X_A|X_C, X_B] = \mathbb{P}[X_A|X_C] \otimes \mathbb{I}[X_B] .$$

Proof sketch: We have by Bayes theorem

$$\mathbb{P}[X_A, X_B|X_C] = \langle \mathbb{P}[X_A|X_C, X_B], \mathbb{P}[X_B|X_C] \rangle_{[X_A, X_B, X_C]}$$

and compare with the conditional independence equation

$$\mathbb{P}[X_A, X_B|X_C] = \langle \mathbb{P}[X_A|X_C], \mathbb{P}[X_B|X_C] \rangle_{[X_A, X_B, X_C]} .$$

Usage for decompositions: We can drop some conditioned variables and effectively smallen the tensor order in the tensors $\mathbb{P}[X_k|X_{[k]}]$.

Example: Markov Chains

Let us model the weather in a town on all days of a year by variables $X_{[365]}$ and assume that for each $k \in [365]$

- ▶ Three states $x_k = 0$ (sunny), $x_k = 1$ (cloudy), $x_k = 2$ (rainy)
- ▶ The weather on the day $k + 1$ conditioned on the weather on the day k is

$$\mathbb{P}[X_{k+1}|X_k] = \begin{pmatrix} 0.70 & 0.25 & 0.05 \\ 0.35 & 0.40 & 0.25 \\ 0.10 & 0.40 & 0.50 \end{pmatrix} [X_{k+1}, X_k]$$

- ▶ The weather on the day $k + 1$ does not depend on the weather before day k given the weather before day k :

$$(X_{k+1} \perp X_{[k]}) \mid X_k$$

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- ▶ The weather on the day $k + 1$ does not depend on the weather before day k given the weather before day k :

$$(X_{k+1} \perp X_{[k]}) \mid X_k$$

This is an example of a Markov chain: Variables depend only on their predecessors.

Example: Markov Chains

Theorem (Markov Chain)

Let there be a set of variables X_k where $k \in [d]$. When X_k is independent of $X_{[k-1]}$ given X_{k-1} (the Markov Property), then

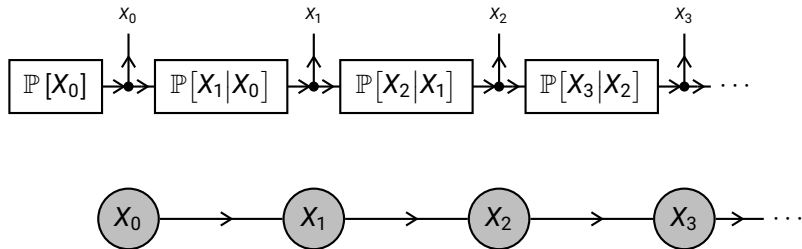
$$\mathbb{P}[X_{[d]}] = \langle \mathbb{P}[X_k | X_{k-1}] : k \in [d] \rangle_{[X_{[d]}]} .$$

Proof sketch: We use the chain decomposition and exploit the independence assumptions to get

$$\mathbb{P}[X_k | X_{[k]}] = \mathbb{P}[X_k | X_{k-1}] \otimes \mathbb{I}[X_{[k-1]}] .$$

Tensor network decompositions of markov chains

We depict this by



Can we find a more generic correspondence between probability decompositions and conditional independence statements?

Outline

- ▶ Independence of variables
- ▶ Conditional independence
- ▶ Decompositions given independence
- ▶ **Bayesian networks**
- ▶ Markov networks

Independence assumptions of Bayesian networks

Towards decomposing a distribution in a more generic way take the steps:

- ▶ We choose an order and enumerate the variables by $[d]$.
- ▶ To each $k \in [d]$ we choose a set $\text{Pa}(k)$ of **parent nodes** in $[k] = \{0, \dots, k-1\}$ such that

$$(X_k \perp X_{[k] \setminus \text{Pa}(k)}) \mid X_{\text{Pa}(k)}$$

Example: Markov chains are the special case of $\text{Pa}(k) = \{k-1\}$.

Factorization into Bayesian networks

Theorem

When for each $k \in [d]$

$$(X_k \perp X_{[k] \setminus \text{Pa}(k)}) \mid X_{\text{Pa}(k)}$$

then

$$\mathbb{P}[X_{[d]}] = \langle \{\mathbb{P}[X_k | X_{\text{Pa}(k)}] : k \in [d]\} \rangle_{[X_{[d]}]} .$$

Proof sketch: As in the proof of the Markov chain factorization theorem, we use the chain factorization and exploit the conditional independences to show that

$$\mathbb{P}[X_k | X_{[k]}] = \mathbb{P}[X_k | X_{\text{Pa}(k)}] \otimes \mathbb{I}[X_{[k] \setminus \text{Pa}(k)}] .$$

Example: At the Dentist

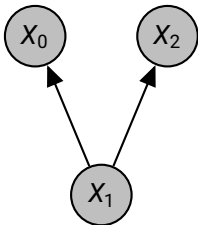
- Choose the order X_1, X_0, X_2
- For both X_0 and X_2 we take X_1 as single parent, that is

$$\text{Pa}(0) = \{1\} \quad \text{and} \quad \text{Pa}(2) = \{1\}$$

and have the for the Bayesian network necessary conditional independencies (shown before)

$$(X_0 \perp X_2) \mid X_1 \quad \text{and} \quad (X_2 \perp X_0) \mid X_1$$

The Toothache



$$\mathbb{P}[X_0, X_1, X_2] = \mathbb{P}[X_0|X_1] \mathbb{P}[X_2|X_1] \mathbb{P}[X_1]$$

A graphical representation of the joint probability distribution factorization. It shows three rectangular boxes. The left box contains $\mathbb{P}[X_0|X_1]$ and has an upward arrow labeled X_0 . The right box contains $\mathbb{P}[X_2|X_1]$ and has an upward arrow labeled X_2 . Below these two boxes is a node with an upward arrow labeled X_1 . Two curved arrows point from this bottom node to the bottom of each of the two boxes above it, representing the conditioning of X_0 and X_2 on X_1 .

Example: Hidden Markov Models

Extend the weather example: We observe the weather E_k , but not the hidden driving variable:

- Atmospheric pressure X_k taking states $x_k = 0$ (high pressure) or $x_k = 1$ (low pressure)

Transition of the pressure variable:

$$\begin{pmatrix} 0.80 & 0.20 \\ 0.30 & 0.70 \end{pmatrix} [X_{k+1}, X_k]$$

Probability of whether given the

$$\begin{pmatrix} 0.70 & 0.25 & 0.05 \\ 0.10 & 0.40 & 0.50 \end{pmatrix} [E_k, X_k]$$

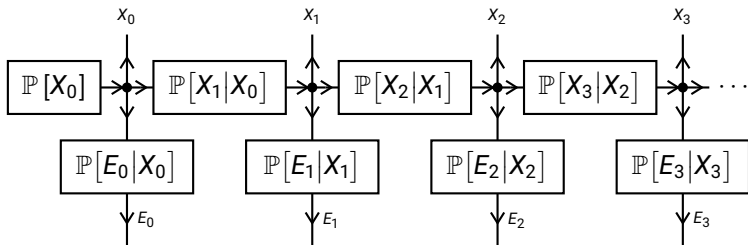
This is an example of Hidden Markov Models extending Markov Chains by limited observation E_k of the variables X_k .

Example: Hidden Markov Models

The independence assumptions of a Hidden Markov Model are for any $k \in [d]$

- ▶ X_{k+1} is independent of $X_{[k]}$ and $E_{[k]}$ given X_k
- ▶ E_k is independent of all other variables given X_k

The independence assumptions are exploited in the decomposition



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Undirected Graphical Models: Markov Networks

Definition (Markov Networks)

Let $\{\tau^e[X_e] : e \in \mathcal{E}\}$ be a Tensor Network on a hypergraph $\mathcal{G}$. The associated Markov Network is the probability distribution of $\{X_v : v \in \mathcal{V}\}$ defined by

$$\mathbb{P}[X_{\mathcal{V}}] = \frac{\langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[X_{\mathcal{V}}]}}{\langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[\emptyset]}} = \langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[X_{\mathcal{V}}|\emptyset]} .$$

We call the denominator

$$\mathcal{Z}(\mathcal{G}) = \langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[\emptyset]}$$

the **partition function** of the Markov Network.

Markov network independencies

Theorem

A positive distribution $\mathbb{P}[X_{\mathcal{V}}]$ has a representation as a Markov network on a hypergraph $\mathcal{G}$, if and only if for any disjoint subsets $\mathcal{U}, \mathcal{W}, \mathcal{I} \subset \mathcal{V}$ for which $\mathcal{I}$ separates $\mathcal{U}$ from $\mathcal{W}$

$$(X_{\mathcal{U}} \perp X_{\mathcal{W}}) \mid X_{\mathcal{I}}.$$

Proof sketch: This is the Hammersley Clifford factorization theorem, which proof is out of scope in this course.